

same? So does not innovation take precedence here?

There are two things here—innovation and invention. Invention refers to unique and new creations, often protected by patents or released into the public domain, while innovation involves improving methodologies, marketing strategies, and products. You can have the same product which can be marketed by different people in different ways, or include innovative



processes for development, engaging with clients, or improvements in the existing designs. We focus on innovation, particularly in making products market-ready, addressing customer needs effectively. While invention is valued in academic contexts like PhD theses, innovation is crucial for addressing customer needs and reaching end-users effectively. So innovation is at the core of our approach to entrepreneurship and business development.

How do you support these individuals who come up with their start-ups, with respect to the challenges they face?

Start-ups in India, especially in hardware, face numerous challenges, including access to testing and certification facilities, as well as pilot manufacturing capabilities. These resources require a significant investment of both time and

money, which start-ups often lack. To address this, we aim to create a supportive network that provides access to funding, connections, and knowledge in areas such as legal and marketing support. Our ecosystem, comprising various centres with expertise in different domains, facilitates this support. By connecting start-ups with the necessary resources and expertise, we strive to navigate these challenges and enable their growth.

Are your facilities equipped for large-scale manufacturing or just prototyping?

There are three layers to consider: prototyping, pilot testing, and scale-up. Prototyping requires large support certification. Pilot testing facilities for small-volume production are lacking domestically. Currently, for small-scale volume testing, few facilities exist, particularly for sensors. Many hardware projects relocate due to better systems abroad. A well-connected network is crucial for timely product readiness and certifications.

Which are the current market segments that SMEs or start-ups should start in order to get that go-ahead or give that push to the ecosystem?

In the IoT industry, hardware manufacturing is limited, with most focus on software development. However, there is an opportunity for hardware innovation, such as board designs and optimising data processing for energy efficiency and speed. India has a shortage of hardware engineering skills, particularly in firmware and compiler design. Accessibility to technical skills and supply chain issues hinder entry into this field. Developing valuable IPs and making VLSI and electronics careers more accessible is crucial for India's growth in hardware innovation. **EFY**

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